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# POSTER ABSTRACTS

## 723.ALLOGENEIC TRANSPLANTATION: LONG-TERM FOLLOW-UP AND DISEASE RECURRENCE

### Phase II Trial of Venetoclax (Ven) in Combination with Azacitidine (AZA) As Maintenance Therapy for High-Risk Acute Leukemia Following Allogeneic Stem Cell Transplantation (SCT)

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**Abstract Introduction:** Allogeneic SCT remains the only potentially curative therapy for acute leukemia patients but with inferior outcomes for high-risk patients. Prior data on maintenance therapies, including AZA, have yielded conflicting results on its efficacy to improve outcomes. Considering the clinical efficacy of VEN in acute myeloid and lymphoid leukemia (AML/ALL) with its known tolerability when used with AZA, we thought VEN and AZA might be an attractive maintenance strategy to improve outcomes after SCT.

**Methods:** This is a single center, open label, Phase II clinical trial to assess the safety and efficacy of VEN with AZA as maintenance therapy after SCT. Inclusion criteria was 1) high risk AML or B cell and T cell ALL or mixed phenotype leukemia, 2) absolute neutrophil count >/= 1.0 x $10^9$/L and platelet >/= 30 x $10^9$/L without any support, 3) absence of active graft versus host disease (GvHD) and infection, 4) absence of minimal residual disease at enrollment.

Treatment consisted of up to 12 cycles of VEN 100 mg on Days 1-7 with AZA 32 mg/m $^2$/day IV/SC on Days 1-5 every 28 days. VEN dosing was adjusted for patients on concomitant CYP3A inhibitors or those with severe cytopenias. After the first 11 patients enrolled, due to myelosuppression observed, VEN dose was changed to 50 mg on days 1-7 if patients were on posaconazole. Treatment delays, up to 70 days, were allowed by toxicity.

The primary objectives are relapse-free survival (RFS) after initiation of VEN and AZA and the safety and tolerability of the combination. The study has been ongoing.

**Results:** The study enrolled a total of 30 patients (27 AML, 3 ALL). Demographics across all AML patients included mean age 52 years (range: 40-74), 67% males, 15% second SCT, 48% in first complete remission (CR1) at SCT, 41% adverse risk by European Leukemia Net classification. The majority (63%) received a myeloablative intensity conditioning regimen using intravenous busulfan. The three ALL patients were younger with mean age of 35 and all were in CR1 at SCT.

The median time to start the maintenance treatment was 70.5 days (range, 42-103). As of the data cutoff date of 13 July 2022, patients completed a median of 4 cycles (range, 1-12) of study treatment with 9 patients (30%) remaining on study treatment. The primary reasons for study treatment discontinuation among 21 patients were complications like myelosuppresssion, infections and GvHD in 11, relapse in 6 and completion of 12 cycles of treatment in 4.

The median follow-up time was 8.67 months (95% CI, 7.23 to 13.6).The median RFS and OS were not reached. The estimates of RFS and OS at 1-year were 69.2% (95% CI, 52.1% to 91.8%) and 90.2% (95%CI, 78% to 100%), respectively, after the initiation of VEN and AZA maintenance (Figure 1&2).

Assuming a Bayesian Poisson-Gamma Model with a Gamma (1,1) prior; the mean number of grade≥3 treatment related adverse events (AEs) per patient was estimated. The mean number of AEs were 1.87 (range, 0-8) for decreased white blood cell count; 1.68 (range, 0-7) for decreased absolute neutrophil count; 1.06 (range, 0-6) for decreased platelet count; 0.55 (range, 0-6) for anemia; 0.13 (range, 0-1) for pneumonia and 0.06 (range 0-1) for each event of febrile neutropenia,

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maculopapular rash, respiratory failure and secondary graft failure (GF). The secondary GF was observed in one patient after the 4th cycle of treatment leading to discontinuation of study treatment. No patient died to due to treatment related AEs. Due to frequent and prolonged myelosuppression observed, 9 patients (30%) required dose reduction of VEN; mostly right after the first cycle of treatment (6 of 9 patients). After the protocol amendment for VEN to be given as 50 mg on days 1-7 on posaconazole, the requirement for dose changes decreased though still some patients required lower doses of VEN (as 30 mg on days 1-5) on subsequent cycles.

**Conclusion:** Post-SCT maintenance therapy with VEN/AZA was safe and tolerable with favorable results in patients with high-risk AML and ALL. The majority of reported AEs was related to myelosuppression which is a known complication of VEN. Many patients required dose modification of VEN and treatment delays between cycles, highlighting the importance of monitoring these patients very closely. On the other hand, the observed RFS and OS data were highly encouraging compared to the historical outcomes for this high-risk group of patients although longer follow-up is needed. The study has been ongoing.

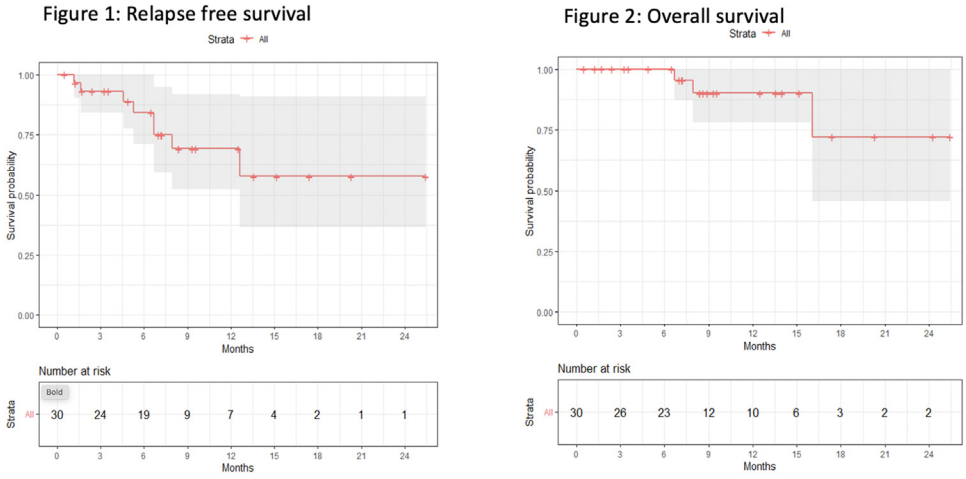



**Figure 1.**

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**OffLabel Disclosure:** Venetoclax and Azacitidine used in vombination after the transplant

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